Completing the Square

Algebra · Edexcel IGCSE Mathematics A (4MA1) · Year 10 · worksheet

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Name	Class	Date

Answer ALL questions.

Write your answers in the spaces provided. You must write down all the stages in your working.

Suggested time: 68 minutes · Total marks: 68

KEY RULES

Form	$x^2 + bx + c = \left(x + \frac{b}{2}\right)^2 - \left(\frac{b}{2}\right)^2 + c$
Step 1	Halve the coefficient of x
If $a \neq 1$	Take a out of the x^2 and x terms first: $a(x^2 + \ldots)$
Solving	Complete the square, then rearrange: $(x + p)^2 = k$ gives $x = -p \pm \sqrt{k}$
Turning point	$a(x + p)^2 + q$ has turning point $(-p,\ q)$
Min / max	Minimum value q if $a > 0$; maximum value q if $a < 0$

WORKED EXAMPLE 1

[2 marks]

Write $x^2 + 6x + 2$ in the form $(x + a)^2 + b$.

Step 1	Halve the 6	$(x + 3)^2$
Step 2	Subtract the extra 3^2 and keep the $+2$	$(x + 3)^2 - 9 + 2$

Answer: $(x + 3)^2 - 7$

WORKED EXAMPLE 2

[3 marks]

Write $2x^2 - 8x + 5$ in the form $a(x + p)^2 + q$.

Step 1	Take out 2 from the x terms	$2(x^2 - 4x) + 5$
Step 2	Complete the square inside	$2[(x - 2)^2 - 4] + 5$
Step 3	Expand the bracket and simplify	$2(x - 2)^2 - 8 + 5$

Answer: $2(x - 2)^2 - 3$

WORKED EXAMPLE 3

[3 marks]

Solve $x^2 - 6x - 3 = 0$ by completing the square. Give your answers to 2 decimal places.

Step 1	Complete the square	$(x - 3)^2 - 9 - 3 = 0$
Step 2	Rearrange	$(x - 3)^2 = 12$
Step 3	Square root both sides	$x = 3 \pm \sqrt{12} = 3 \pm 3.464$

Answer: $x = 6.46$ **or** $x = -0.46$

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Completing the Square

1 Write $x^2 + 4x + 7$ in the form $(x + a)^2 + b$.

(Total for question = 2 marks)

Practice question written for this worksheet (not from a past paper)

2 Write $x^2 - 10x + 40$ in the form $(x + a)^2 + b$.

(Total for question = 2 marks)

Practice question written for this worksheet (not from a past paper)

3 Write $x^2 + 8x - 3$ in the form $(x + a)^2 + b$.

(Total for question = 2 marks)

Practice question written for this worksheet (not from a past paper)

4 Write $x^2 - 5x + 1$ in the form $(x + a)^2 + b$.

(Total for question = 2 marks)

Practice question written for this worksheet (not from a past paper)

5 Solve $x^2 + 6x - 2 = 0$ by completing the square.

Give your answers in the form $a \pm \sqrt{b}$.

(Total for question = 3 marks)

Practice question written for this worksheet (not from a past paper)

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6 Write $3x^2 + 12x + 5$ in the form $a(x + p)^2 + q$.

(Total for question = 3 marks)

Practice question written for this worksheet (not from a past paper)

7 Write $x^2 - 6x + 11$ in the form $(x + a)^2 + b$.

Hence write down the minimum value of $x^2 - 6x + 11$ and the value of x where it

(Total for question = 3 marks)

Practice question written for this worksheet (not from a past paper)

8 Express $3x^2 - 6x + 5$ in the form $a(x - b)^2 + c$

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9 Express $5x^2 - 20x + 23$ in the form $a(x - b)^2 + c$ where a, b and c are integers.

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10 (a) Write $3x^2 - 12x + 7$ in the form $a(x + b)^2 + c$

(3)

The line **L** is the line of symmetry of the curve with equation $y = 3x^2 - 12x + 7$

(b) Using your answer to part (a) or otherwise, write down an equation of **L**.

(1)

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11 **20** (a) Express $2x^2 - 11x + 9$ in the form $a(x - b)^2 - c$ where a, b and c are numbers to be found.

.....

(3)

The curve **C** has equation $y = 2(x - 3)^2 - 11(x - 3) + 9$

The point P is the minimum point on **C**

(b) Find the coordinates of P

(..... , ...)

(2)

(Total for Question 20 is 5 marks)

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12 **23** (a) Express $2x^2 - 12x + 3$ in the form $a(x + b)^2 + c$ where a, b and c are integers.

.....

(3)

The curve **C** has equation $y = 2(x + 4)^2 - 12(x + 4) + 3$

The point M is the minimum point on **C**

(b) Find the coordinates of M

(..... , ...)

(2)

(Total for Question 23 is 5 marks)

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13 (a) Express $10 - 24x - 3x^2$ in the form $a - b(x + c)^2$ where a, b and c are integers.

.....

(3)

T is the curve with equation $y = 10 - 24(x - 1) - 3(x - 1)^2$
The point P is the maximum point on **T**

b) Use your answer to part (a) to find the coordinates of P

(..... , ...)

(1)

Write down the equation for the line of symmetry of **T**

.....

(1)

(Total for Question 24 is 5 marks)

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14 (a) Write $7 + 12x - 3x^2$ in the form $a + b(x + c)^2$ where a, b and c are integers.

.....

(4)

The curve **C** has equation $y = 7 + 12x - 3x^2$
The point A is the turning point on **C**.

(b) Using your answer to part (a), write down the coordinates of A.

(.....,.....)

(1)

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CHALLENGE QUESTIONS

15 (a) Express $7 - 4x - x^2$ in the form $p - (x + q)^2$ where p and q are constants.

.....

(2)

(b) Use your answer to part (a) to solve the equation $7 - 4(y + 3) - (y + 3)^2 = 0$

Give your solutions in the form $e \pm \sqrt{f}$ where e and f are integers.

.....

(3)

The curve **C** has equation $y = 3 - 5(x + 1)^2$
The point A is the maximum point on **C**.

(c) Write down the coordinates of A.

(.....,.....)

(1)

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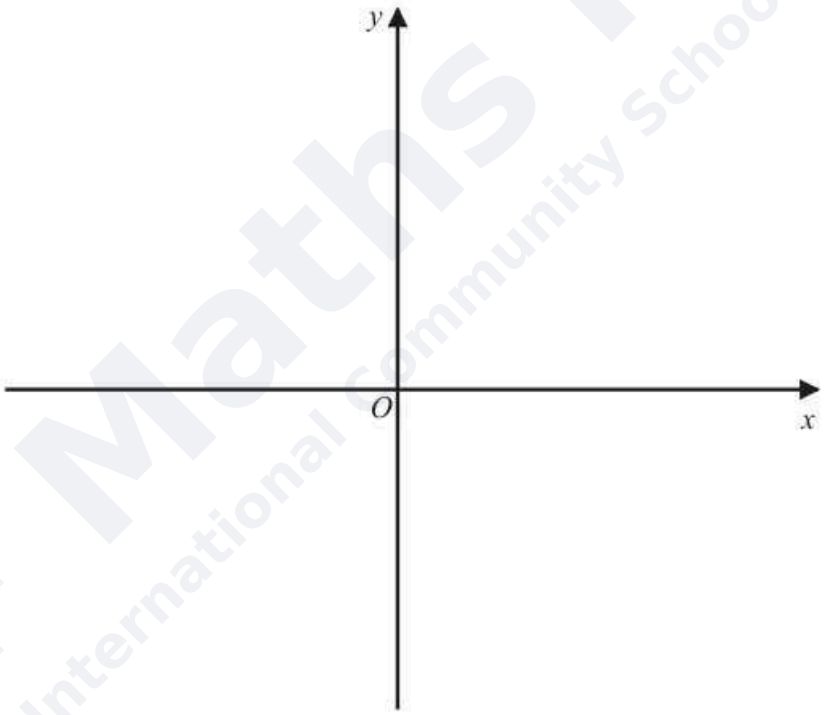
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16 The curve **C** has equation $y = 4(x - 1)^2 - a$ where $a > 4$

Using the axes below, sketch the curve **C**.
On your sketch show clearly, in terms of a,

(i) the coordinates of any points of intersection of **C** with the coordinate axes,

(ii) the coordinates of the turning point.



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17

23 The curve **C** has equation $y = x^2 - 8x - 9$
The straight line **L** has equation $y = k$ where k is an integer.

C and **L** intersect at the points A and B

The coordinates of point A are (p, k)
The coordinates of point B are (q, k)

Given that $p - q = 14$

find the value of k
Show clear algebraic working.

$k =$ .....

(Total for Question 23 is 5 marks)

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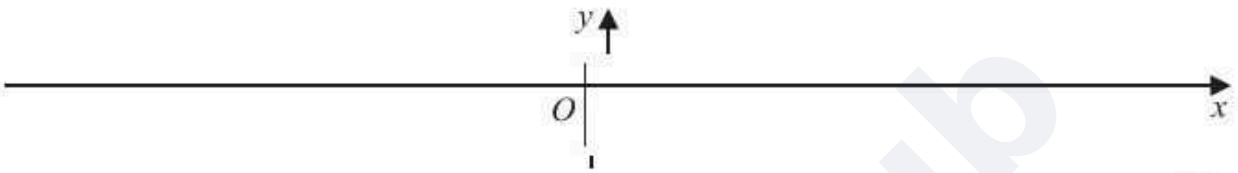
18 (a) Write $28+24x-6x^2$ in the form $a-b(x-c)^2$ where a, b and c are integers.

.....

(3)

(b) On the axes below, sketch the graph of $y=28+24x-6x^2$

Show clearly the coordinates of the turning point and the coordinates of the point of intersection of the graph with the y-axis.



(3)

(Total for Question 25 is 6 marks)

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TOTAL FOR PAPER IS 68 MARKS

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